One-dimensional Turbulence Model (notes)

Sienna White

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Governing equations for flow

We first discuss the derivation of the equations used for our one-dimensional water column model.

Reynolds-averaged Navier Stokes

We start with the Reynolds-Averaged Navier Stokes, which reads:

$$\frac{\partial \overline{u_i}}{\partial t} + \overline{u_j}\frac{\partial \overline{u_i}}{\partial x_j} = \frac{\partial}{\partial x_k}\overline{u_k u_i} - \frac{1}{\rho_0}\frac{\partial \overline{p}}{\partial x_i} - g\delta_{i3} + \nu\frac{\partial^2}{\partial x_j \partial x_j}\overline{u_i} \tag{1}$$

where flow is considered to be incompressible (solenoidal),

$$\frac{\partial u_i}{\partial x_i} = 0. \tag{2}$$

This equation in the $\hat{e_1}$ direction, expanded, reads

$$\frac{\partial \overline{u}}{\partial t} + \overline{u}\frac{\partial \overline{u}}{\partial x} + \overline{v}\frac{\partial \overline{u}}{\partial y} + \overline{w}\frac{\partial \overline{u}}{\partial z} = \frac{\partial}{\partial x}\overline{u'u'} + \frac{\partial}{\partial y}\overline{v'u'} + \frac{\partial}{\partial z}\overline{w'u'} - \frac{1}{\rho_0}\frac{\partial \overline{p}}{\partial x} + \nu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right).$$

We neglect advection and gradients in the x and y directions and take $w = 0$ (established by the boundary layer approxmiation),

$$\frac{\partial \overline{u}}{\partial t} + \cancel{\overline{u}\frac{\partial \overline{u}}{\partial x}} + \cancel{\overline{v}\frac{\partial \overline{u}}{\partial y}} + \cancel{\overline{w}\frac{\partial \overline{u}}{\partial z}} = \cancel{\frac{\partial}{\partial x}\overline{u'u'}} + \cancel{\frac{\partial}{\partial y}\overline{v'u'}} + \frac{\partial}{\partial z}\overline{w'u'} - \frac{1}{\rho_0}\frac{\partial \overline{p}}{\partial x} + \nu\left(\cancel{\frac{\partial^2 u}{\partial x^2}} + \cancel{\frac{\partial^2 u}{\partial y^2}} + \frac{\partial^2 u}{\partial z^2}\right)$$

We now have a simplified equation for Reynolds-averaged flow in the $\hat{e_1}$ direction.

$$\frac{\partial \overline{u}}{\partial t} = \frac{\partial}{\partial z}\overline{w'u'} - \frac{1}{\rho_0}\frac{\partial \overline{p}}{\partial x} + \nu\frac{\partial^2 u}{\partial z^2}. \tag{3}$$

Next, we apply the Boussinesq assumption, which allows us to re-write the Reynolds Stress as a function of the mean velocity gradient. Note that the k term drops out as our Reynolds stress is off-diagonal,

$$\overline{w'u'} = \mu_t\left(\frac{\partial \overline{u}}{\partial z} + \frac{\partial \overline{w}}{\partial x}\right) - \cancel{\frac{2}{3}k\delta_{13}} = \mu_t\left(\frac{\partial \overline{u}}{\partial z} + \cancel{\frac{\partial \overline{w}}{\partial x}}\right) = \mu_t\frac{\partial \overline{u}}{\partial z}$$

We plug our expression for the Reynold's stress into Eq. 3,

$$\frac{\partial \overline{u}}{\partial t} = \frac{\partial}{\partial z}\left(\mu_t \frac{\partial \overline{u}}{\partial z}\right) - \frac{1}{\rho_0}\frac{\partial \overline{p}}{\partial x} + \nu \frac{\partial^2 \overline{u}}{\partial z^2}$$

$$\frac{\partial \overline{u}}{\partial t} = \mu_t \frac{\partial^2 \overline{u}}{\partial z^2} - \frac{1}{\rho_0}\frac{\partial \overline{p}}{\partial x} + \nu \frac{\partial^2 \overline{u}}{\partial z^2}$$

We can group the turbulent and molecular viscosity terms,

$$\frac{\partial \overline{u}}{\partial t} = -\frac{1}{\rho_0}\frac{\partial \overline{p}}{\partial x} + (\nu_t + \nu)\frac{\partial^2 \overline{u}}{\partial z^2}$$

however, since we assume that turbulent viscosity is far greater than molecular viscosity,

$$\nu_t >> \nu$$

we often neglect molecular viscosity, leaving us with

$$\frac{\partial \overline{u}}{\partial t} = -\frac{1}{\rho_0}\frac{\partial \overline{p}}{\partial x} + \nu_t \frac{\partial^2 \overline{u}}{\partial z^2}.$$

After applying the Boussinesq assumption, we've escaped the need to resolve the Reynold's Stress, however, in discretizing this equation, we still need to approximate this turbulent viscosity ν_t. Hence the need for a closure scheme.

Turbulence closure

Similarity theory tells us that ν_t, which has units of [length^2 time^{-1}] can be scaled by a length scale times a velocity scale,

$$\nu_t \approx \ell_T \, U_T$$

Here I am using the subscript T to denote these quanitities being *turbulent scales.* We estimate the velocity scale using the square root of the total turbulent kinetic energy,

$$\text{TKE} = \text{tr}(\overline{u_i u_j}) = (\overline{u'^2} + \overline{v'^2} + \overline{w'^2})$$

$$q = \sqrt{\overline{u'^2} + \overline{v'^2} + \overline{w'^2}}$$

$$\nu_t \approx \ell_T \, q$$

Again, ν_t is a function $\nu_t(z,t)$ that dissipates energy from the mean velocity (due to subscale turbulence). We use the Mellor-Yamada 2.5 scheme to parametrize ν_t.

Discretization

Velocity

Our governing equation for flow is given by,

$$\frac{\partial u}{\partial t} = -\frac{1}{\rho}\frac{\partial p}{\partial z} + \nu_t \frac{\partial^2 u}{\partial z^2} \tag{4}$$

We discretize this equation in time and space. We break down the discretization of Equation 4 by the left hand (unsteadiness) and right hand (pressure + diffusion) sides. The right-hand side is trickiest. We use an implicit scheme for time and a central difference scheme for space.

$$\frac{\partial}{\partial z}\left(\nu_t \frac{\partial u}{\partial z}\right) \approx \frac{1}{\Delta z}\left(\left(\nu_t \frac{\partial u}{\partial z}\right)_{i+1/2} - \left(\nu_t \frac{\partial u}{\partial z}\right)_{i-1/2}\right)$$

evaluating velocity u at $i+1$, $i-1$. Dropping the subscript on ν_t, we rewrite this as

$$= \frac{1}{\Delta z}\left(\nu_{i+1/2} \frac{\partial u}{\partial z}_{i+1} - \nu_{i-1/2} \frac{\partial u}{\partial z}_{i-1}\right)$$

and discretizing the remaining gradients gives us,

$$= \frac{1}{\Delta z^2}\left(\nu_{i+1/2}\left(u_{i+1} - u_i\right) - \nu_{i-1/2}\left(u_i - u_{i-1}\right)\right)$$

$$= \frac{1}{\Delta z^2}\left(\nu_{i+1/2} u_{i+1} - (\nu_{i+1/2} + \nu_{i-1/2}) u_i + \nu_{i-1/2} u_{i-1}\right)$$

Putting it all together, we get

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = -Px_i^n + \frac{1}{\Delta z^2}\left(\nu_{i+1/2} u_{i+1} - (\nu_{i+1/2} + \nu_{i-1/2}) u_i + \nu_{i-1/2} u_{i-1}\right) \tag{5}$$

$$u_i^{n+1} - u_i^n = -\Delta t P x_i^n + \frac{\Delta t}{\Delta z^2}\left(\nu_{i+1/2} u_{i+1} - (\nu_{i+1/2} + \nu_{i-1/2}) u_i + \nu_{i-1/2} u_{i-1}\right)$$

Rearrange, grouping terms by time step (past on left, future on the right),

$$u_i^n - \Delta t P x_i^n = u_i^{n+1} - \frac{\Delta t}{\Delta z^2}\left(\nu_{i+1/2} u_{i+1} - (\nu_{i+1/2} + \nu_{i-1/2}) u_i + \nu_{i-1/2} u_{i-1}\right)$$

Wind stress

Consider the partial discretization,

$$u_i^{n+1} - u_i^n = -\Delta t P x_i^n = \frac{\Delta t}{\Delta z}\left(\nu_{i+1/2}\frac{\partial u}{\partial z}_{i+1} - \nu_{i-1/2}\frac{\partial u}{\partial z}_{i-1}\right)^{n+1}$$

When there is wind stress at the surface, we expect

$$\frac{\partial u}{\partial z} = \frac{u_*}{\kappa \Delta z} = \frac{W}{\Delta z}$$

$$u_i^{n+1} - u_i^n = -\Delta t P x_i^n + \frac{\Delta t}{\Delta z}\left(\nu_{i+1/2} W - \nu_{i-1/2}\frac{\partial u}{\partial z}_{i-1}\right)^{n+1}$$

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = -P x_i^n + \frac{1}{\Delta z^2}\left(\nu_{i+1/2} W - \nu_{i-1/2}\left(u_i - u_{i-1}\right)\right)$$

$$u_n^n - \Delta t P x_i^n = u_i^{n+1} - \frac{\Delta t}{\Delta z^2}\left(\nu_{i+1/2} W - \nu_{i-1/2}\left(u_i - u_{i-1}\right)\right)^{n+1}$$

$$u_N^n - \Delta t P x_N^n = u_i^{n+1} - \frac{\Delta t}{\Delta z^2}\left(\nu_{N+1/2} W - \nu_{N-1/2} u_i + \nu_{N-1/2} u_{N-1}\right)^{n+1}$$

Advection-Diffusion Equation for Algae

Our governing equation for algae is given by,

$$\frac{\partial C}{\partial t} + w_s \frac{\partial C}{\partial z} = \frac{\partial}{\partial z}\left(\kappa_t \frac{\partial C}{\partial z}\right) + \Gamma C \tag{6}$$

where C is the concentration of algae (scalar), Γ is the source/loss term, and w_s is the sinking or swimming speed of the algae: w_s can be positive (*Microcystis*) or negative (phytoplankton).

We break down the discretization of Equation 6 by the left hand (time derivative + advection) and right hand (diffusion) sides. The right-hand side is trickiest.

We rewrite our expression as,

$$\frac{\partial}{\partial z}\left(\kappa_t \frac{\partial C}{\partial z}\right) \approx \frac{1}{\Delta z}\left(\left(\kappa_t \frac{\partial C}{\partial z}\right)_{i+1/2} - \left(\kappa_t \frac{\partial C}{\partial z}\right)_{i-1/2}\right)$$

evaluating C at $i+1$, $i-1$ (this is where I will now drop the subscript on κ_t),

$$= \frac{1}{\Delta z}\left(\kappa_{i+1/2}\frac{\partial C}{\partial z}_{i+1} - \kappa_{i-1/2}\frac{\partial C}{\partial z}_{i-1}\right).$$

Discretizing diffusion

We first evaluate the diffusion term in space (neglecting Γ for now) using a staggered with with central differencing. Note here that diffusion is given by κ_t, which respresents turbulent diffusion (a quantity we will solve for in the turbulence closure). Because the subscripts get messy, I'll drop the subscript t, but all diffusion in this equation refers to κ_t.

For compactness, we evaluate $\kappa_t \frac{\partial C}{\partial z}$ at $i + \frac{1}{2}$, $i - \frac{1}{2}$. See the below diagram for a visual representation of this stencil.

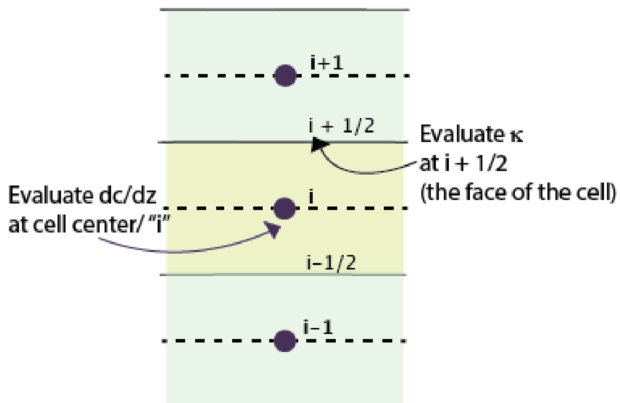

Figure 1: Staggered stencil for κ_t & C

We rewrite our expression as,

$$\frac{\partial}{\partial z}\left(\kappa_t \frac{\partial C}{\partial z}\right) \approx \frac{1}{\Delta z}\left(\left(\kappa_t \frac{\partial C}{\partial z}\right)_{i+1/2} - \left(\kappa_t \frac{\partial C}{\partial z}\right)_{i-1/2}\right)$$

evaluating C at $i+1$, $i-1$ (this is where I will now drop the subscript on κ_t),

$$= \frac{1}{\Delta z}\left(\kappa_{i+1/2} \frac{\partial C}{\partial z}_{i+1} - \kappa_{i-1/2} \frac{\partial C}{\partial z}_{i-1}\right).$$

Note that evenutally we will represent these $\frac{1}{2}$ stencil points for κ_t as the average of the two neighboring cells,

$$\kappa_{i+1/2} \approx \frac{\kappa_{i+1} + \kappa_i}{2}$$

$$\kappa_{i-1/2} \approx \frac{\kappa_{i-1} + \kappa_i}{2}$$

For now though, just to minimize the amount of terms we juggle in the algebra, I'll leave these written as $\kappa_{i-/+1/2}$.

Now we discretize the graident of C,

$$\left(\frac{\partial C}{\partial z}\right)_{i+1} \approx \frac{C_{i+1} - C_i}{\Delta z}$$

$$\left(\frac{\partial C}{\partial z}\right)_{i-1} \approx \frac{C_i - C_{i-1}}{\Delta z}$$

Putting these together,

$$\frac{1}{\Delta z}\left(\kappa_{i+1/2}\frac{C_{i+1} - C_i}{\Delta z} - \kappa_{i-1/2}\frac{C_i - C_{i-1}}{\Delta z}\right)$$

Factor according to C,

$$= C_{i+1}\frac{1}{\Delta z^2}\left(\kappa_{i+1/2}\right) + C_i\frac{1}{\Delta z^2}\left(-\kappa_{i+1/2} - \kappa_{i-1/2}\right) + C_{i-1}\frac{1}{\Delta z^2}\left(\kappa_{i-1/2}\right) \tag{7}$$

Note that this expression (Eq 7) for diffusion will not change depending on the sign of w_s.

Discretizing advection

Discretizing is advection is trickier as we employ upwinding that depends on the direction of swimming speed (eg, sinking particles ($w_s < 0$) vs. floating particles ($w_s > 0$)). Note that the nature of upwinding with the "algae advection" term changes depending on the sign of w_s. This gets confusing very easily (see diagram).

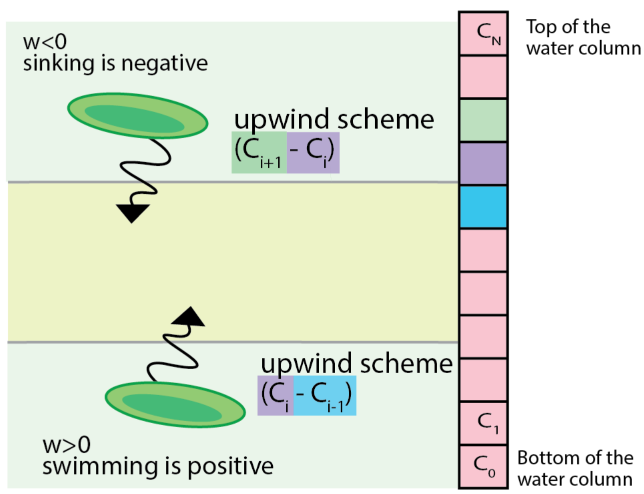

Figure 2: Upwinding discretization for plankton advection depends on the direction of w_s (sinking or swimming)

Discretizing in time

Next we discretize our in time. In order to prevent confusion, let me note that w_s is a positive quantity no matter the direction of the particle. If the particle sinks, we write this equation as,

$$\frac{\partial C}{\partial t} - w_s \frac{\partial C}{\partial z} = \frac{\partial}{\partial z}\left(\kappa_t \frac{\partial C}{\partial z}\right) + \Gamma C \tag{8}$$

and if the algae floats, we write the equation as

$$\frac{\partial C}{\partial t} + w_s \frac{\partial C}{\partial z} = \frac{\partial}{\partial z}\left(\kappa_t \frac{\partial C}{\partial z}\right) + \Gamma C \tag{9}$$

Thus, in our code, we must note the direction of w_s in addition to its magnitude.

For discretization, we use an implicit scheme with upwinding.

If $w_s > 0$ (Eq. 9) such as in the case of Microcystis, where the algae are very-barely-buoyant,

$$\frac{C_i^{n+1} - C_i^n}{\Delta t} + w_s \frac{C_i^{n+1} - C_{i-1}^{n+1}}{\Delta z} = \left(\kappa_t \frac{\partial C}{\partial z}\right)^{n+1} + \Gamma C_i^{n+1}$$

In the case of standard diatoms which sink naturally ($w_s < 0$), Eq. 8, we have

$$\frac{C_i^{n+1} - C_i^n}{\Delta t} - w_s \frac{C_{i+1}^{n+1} - C_i^{n+1}}{\Delta z} = \left(\kappa_t \frac{\partial C}{\partial z}\right)^{n+1} + \Gamma C_i^{n+1}.$$

We will solve these individually and discuss our approach for boundary conditions.

Discretized equation when $w_s < 0$

$$\frac{C_i^{n+1} - C_i^n}{\Delta t} - w_s \frac{C_{i+1}^{n+1} - C_i^{n+1}}{\Delta z} = \left(\kappa_t \frac{\partial C}{\partial z}\right)^{n+1} + \Gamma C_i^{n+1}$$

Plug in our expression for diffusion,

$$\frac{C_i^{n+1} - C_i^n}{\Delta t} - w_s \frac{C_{i+1}^{n+1} - C_i^{n+1}}{\Delta z} = \Gamma C_i^{n+1}$$
$$+ C_{i+1} \frac{1}{\Delta z^2}\left(\kappa_{i+1/2}\right) + C_i \frac{1}{\Delta z^2}\left(-\kappa_{i+1/2} - \kappa_{i-1/2}\right) + C_{i-1} \frac{1}{\Delta z^2}\left(\kappa_{i-1/2}\right)$$

Multiply through by Δt and move all C_i^{n+1} terms to the RHS. Designate $\beta = \frac{\Delta t}{\Delta z^2}$.

$$
\begin{aligned}
C_i^n = C_i^{n+1} + \frac{\Delta t w_s}{\Delta z}(C_{i+1}^{n+1} - C_i^{n+1}) - \Gamma C_i^{n+1} \\
-C_{i+1}\beta\left(\kappa_{i+1/2}\right) - C_i\beta\left(-\kappa_{i+1/2} - \kappa_{i-1/2}\right) - C_{i-1}\beta\left(\kappa_{i-1/2}\right)
\end{aligned}
$$

and lastly group by spatial index for C,

$$
\begin{aligned}
C_i^n = C_{i+1}^{n+1}\left(-\frac{\Delta t w_s}{\Delta z} - \beta\kappa_{i+1/2}\right) \\
+C_i^{n+1}\left(1 + \frac{\Delta t w_s}{\Delta z} - \Delta t\Gamma + \beta(\kappa_{i+1/2} + \kappa_{i-1/2})\right) \\
+C_{i-1}^{n+1}\left(-\beta\kappa_{i-1/2}\right)
\end{aligned}
$$

Boundary conditions ($w_s < 0$)

For boundary conditions, we want zero flux at the bed (C_0) and surface (C_N). However, due to the advective flux, we need to consider that no algae should *enter* the surface cell.

$$
\frac{C_N^{n+1} - C_N^n}{\Delta t} - w_s\frac{\cancel{C_{N+1}^{n+1}} - C_N^{n+1}}{\Delta z} = \left(\kappa_t\frac{\partial C}{\partial z}\right)_N^{n+1} + \Gamma C_N^{n+1}
$$

We know that

$$
\frac{C_N - C_{N+1}}{\Delta z} = 0
$$

and plug this into our expression for C_N. Note that we also set the advective flux from the $N+1$ cell to zero.

$$
\begin{aligned}
C_N^n = C_{N+1}^{n+1}\left(\cancel{-\frac{\Delta t w_s}{\Delta z}} - \beta\kappa_{N+1/2}\right) \\
+C_n^{n+1}\left(1 + \frac{\Delta t w_s}{\Delta z} - \Delta t\Gamma + \beta(\kappa_{N+1/2} + \kappa_{N-1/2})\right) \\
+C_{N-1}^{n+1}\left(-\beta\kappa_{N-1/2}\right)
\end{aligned}
$$

giving us the following expression for C_N,

$$
C_N^n = C_N^{n+1}\left(1 + \frac{\Delta t w_s}{\Delta z} - \Delta t\Gamma + \beta\kappa_{N-1/2}\right) + C_{N-1}^{n+1}\left(\beta\kappa_{N-1/2}\right)
$$

In code, we implement this with,

```
# Top-Boundary: no flux for scalars
aA[top] =  -beta/2 * (Kzp[top] + Kzp[top-1])
bA[top] =  1 - gamma[top]*dt + beta/2 * (Kzp[top] + Kzp[top-1]) + wsdtdz
dA[top] = Ap[top]
```

Zero-flux through the bed ($w_s < 0$)

We plug in our expression for $C_0 = C_1$ into our discretized equation, evaluated at C_0. We are not concerned about flux into the bed, and consider diatom settling to be a loss term.

$$\begin{aligned} C_0^n = C_0^{n+1}\left(-\frac{\Delta t w_s}{\Delta z} - \beta\kappa_{1/2}\right) \\ +C_0^{n+1}\left(1 + \frac{\Delta t w_s}{\Delta z} - \Delta t\Gamma + \beta(\kappa_{1/2} + \kappa_{-1/2})\right) \\ +C_{-1}^{n+1}\left(-\beta\kappa_{-1/2}\right) \end{aligned}$$

$$C_0^n = C_0^{n+1}\left(1 - \Delta t\Gamma + \beta\kappa_{-1/2}\right) + C_{-1}^{n+1}\left(-\beta\kappa_{-1/2}\right)$$

In code, we implement this with,

```
# Bottom-Boundary: no flux for scalars
bA[0] =  1 + wsdtdz - (gamma[0]*dt) + beta/2*(Kzp[1] + Kzp[0])
cA[0] =  -wsdtdz -beta/2 * (Kzp[1] + Kzp[0])
dA[0] =  Ap[0]
```

Discretized equation when $w_s > 0$

We now explore the discretization for a plankton with a positive swimming velocity, given in Eq. 9.

$$\frac{C_i^{n+1} - C_i^n}{\Delta t} + w_s\frac{C_i^{n+1} - C_{i-1}^{n+1}}{\Delta z} = \left(\kappa_t\frac{\partial C}{\partial z}\right)^{n+1} + \Gamma C_i^{n+1}$$

We plug in our expression for the Laplacian discretized in space (Eq. 7),

$$C_i^{n+1} - C_i^n + \Delta t w_s \frac{C_i^{n+1} - C_{i-1}^{n+1}}{\Delta z} = \Delta t \Gamma C_i^{n+1} + C_{i+1}\beta\left(\kappa_{i+1/2}\right) + \\ C_i\beta\left(-\kappa_{i+1/2} - \kappa_{i-1/2}\right) + C_{i-1}\beta\left(\kappa_{i-1/2}\right)$$

This all simplifies out to,

$$\begin{aligned} C_i^n = C_{i-1}^{n+1}\left(-\frac{\Delta t w_s}{\Delta z} - \beta(\kappa_{i-1/2}^n)\right) \\ +C_i^{n+1}\left(1 - \Delta t \Gamma_i^n + \frac{\Delta t w_s}{\Delta z} + \beta(\kappa_{i+1/2}^n + \kappa_{i-1/2}^n\right) \\ +C_{i+1}^{n+1}\left(-\beta(\kappa_{i+1/2}\right) \end{aligned} \tag{10}$$

Boundary conditions ($w_s > 0$)

For boundary conditions, we apply the same conditions–

$$C_N = C_{N+1}$$

$$C_0 = C_{-1}$$

Note however that there is no equivalent "settling" term for floating algae. Instead, mass cannot enter or leave at the bed or the surface. This requires us to address the advective flux at both boundaries.

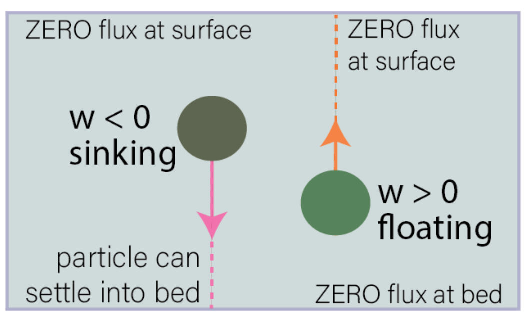

Figure 3: Boundary condition for plankton advection depends on the direction of w_s (sinking or swimming)

Zero-flux through the bed ($w_s > 0$)

Compared to a sinking particle, however, now we must worry about algae advecting from the bed into the water column. We thus cancel out the advective flux coming fron the -1 cell at the bed.

$$\frac{C_0^{n+1} - C_0^n}{\Delta t} + w_s \frac{C_0^1 - \cancel{C_{-1}^{n+1}}}{\Delta z} = \left(\kappa_t \frac{\partial C}{\partial z}\right)^1 + \Gamma C_0^1$$

We apply $C_0 = C_{-1}$ to equation ,

$$\begin{aligned} C_0^n = C_{0-1}^{n+1} &\left(\cancel{-\frac{\Delta t w_s}{\Delta z}} - \beta(\kappa_{-1/2}^n)\right) \\ +C_0^{n+1} \left(1 - \Delta t \Gamma_0^n + \frac{\Delta t w_s}{\Delta z} + \beta(\kappa_{1/2}^n + \kappa_{-1/2}^n)\right) & \\ +C_1^{n+1} &\left(-\beta(\kappa_{1/2}\right) \end{aligned} \tag{11}$$

giving us the final boundary condition of

$$C_0^n = C_0^{n+1} \left(1 - \Delta t \Gamma_0^n + \frac{\Delta t w_s}{\Delta z} + \beta \kappa_{1/2}^n\right) + C_1^{n+1} \left(-\beta \kappa_{1/2}\right)$$

```
# Bottom-Boundary: no flux for scalars
   bA[0] =  1 - (gamma[0]*dt) + beta/2*(Kzp[1] + Kzp[0]) + wsdtdz
   cA[0] = -beta/2 * (Kzp[1] + Kzp[0])
   dA[0] =  Ap[0]
```

Zero-flux through the surface ($w_s > 0$)

Last but not least, we address the surface boundary condition, applying $C_N = C_{N+1}$. Here we also ensure that we impose zero flux through the surface.

$$\frac{C_N^{n+1} - C_N^n}{\Delta t} + w_s \frac{\cancel{C_N^{n+1}} - C_{N-1}^{n+1}}{\Delta z} = \left(\kappa_t \frac{\partial C}{\partial z}\right)^{n+1} + \Gamma C_N^{n+1}$$

$$
\begin{aligned}
C_N^n = C_{N-1}^{n+1}\left(-\frac{\Delta t w_s}{\Delta z} - \beta(\kappa^n_{N-1/2})\right) \\
+ C_N^{n+1}\left(1 - \Delta t \Gamma_N^n + \cancel{\frac{\Delta t w_s}{\Delta z}} + \beta(\kappa^n_{N+1/2} + \kappa^n_{N-1/2}\right) \\
+ C_N^{n+1}\left(-\beta\kappa_{N+1/2}\right)
\end{aligned}
$$

$$
C_N^n = C_{N-1}^{n+1}\left(-\frac{\Delta t w_s}{\Delta z} - \beta(\kappa^n_{N-1/2})\right) + C_N^{n+1}\left(1 - \Delta t \Gamma_N^n + \beta(\kappa^n_{N-1/2}\right)
$$

```
# Top-Boundary: no flux for scalars
aA[top] = -wsdtdz - beta/2 * (Kzp[top] + Kzp[top-1])
bA[top] = 1  - gamma[top]*dt + beta/2 * (Kzp[top] + Kzp[top-1])
dA[top] = Ap[top]
```

Code

In code, to solve this tridiagonal system, we initialize three vectors A, B, and C to represent the diagonals of the matrix.

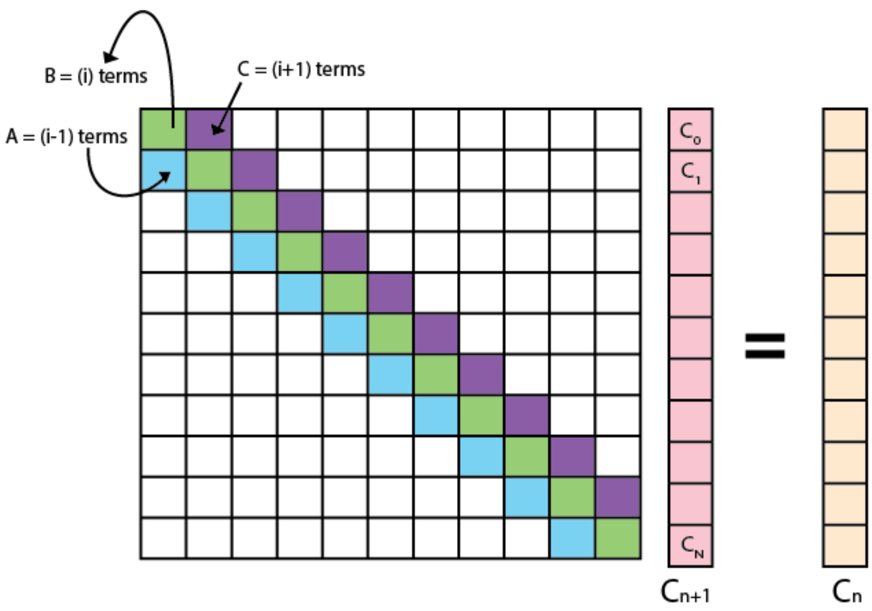

Figure 4: Tridiagonal matrix for implicit scheme

```
# If settling speed is UPWARD (swimming!)
  if ws>0:
      aA[1:top]  = -wsdtdz - beta/2 * (Kzp[0:top-1]+ Kzp[1:top])
      bA[1:top]  = 1 + wsdtdz - gamma[1:top]*dt  \
      + beta/2*(Kzp[2:top+1] + 2*Kzp[1:top] + Kzp[0:top-1])
      cA[1:top]  = -beta/2 * (Kzp[1:top] + Kzp[2:top+1])
      dA = Ap

      # Bottom-Boundary: no flux for scalars
      bA[0] =  1 - (gamma[0]*dt) + beta/2*(Kzp[1] + Kzp[0]) + wsdtdz
      cA[0] = -beta/2 * (Kzp[1] + Kzp[0])
      dA[0] =  Ap[0]

      # Top-Boundary: no flux for scalars
      aA[top] = -wsdtdz - beta/2 * (Kzp[top] + Kzp[top-1])
      bA[top] = 1  - gamma[top]*dt + beta/2 * (Kzp[top] + Kzp[top-1])
      dA[top] = Ap[top]

  # If settling speed is DOWNWARD (sinking!)
  if ws<=0:
      aA[1:top]  = -beta/2 * (Kzp[0:top-1] + Kzp[1:top])
      bA[1:top]  = 1 + wsdtdz - gamma[1:top]*dt  \
      + beta/2*(Kzp[2:top+1] + 2*Kzp[1:top] + Kzp[0:top-1])
      cA[1:top]  = -wsdtdz - beta/2 * (Kzp[1:top] + Kzp[2:top+1])
      dA = Ap

      # Bottom-Boundary: no flux for scalars
      bA[0] =  1 + wsdtdz - (gamma[0]*dt) + beta/2*(Kzp[1] + Kzp[0])
      cA[0] =  -wsdtdz -beta/2 * (Kzp[1] + Kzp[0])
      dA[0] =  Ap[0]

      # Top-Boundary: no flux for scalars
      aA[top] =  -beta/2 * (Kzp[top] + Kzp[top-1])
      bA[top] =  1 - gamma[top]*dt + beta/2 * (Kzp[top] + Kzp[top-1]) + wsdtdz
      dA[top] = Ap[top]
```

Phytoplankton Growth

Now that we have an equation for advection-diffusion of phytoplankton, we can dig into the growth and loss terms. Here, I'll follow the conventions set forth in Huisman's paper.

They discretize the light dependence of phytoplankton growth using a Monod function,

$$p(I) = p_{max} \frac{I}{H + I}$$

H is the half-saturation constant of light-limited growth and p_{max} is the maximum specific production rate for the species. Physically, this means that growth at any given point in the water column is the max-possible growth scaled by a proportion of the amount of available light I and the species' sensitivity to light availability H.

The light available at any point in the water column is a function of the amount of algae above it (self-shading) and background turbidity. They express this using Lambert-Beer's law applied to nonuniform population density distributions:

$$I(z,t) = I_{in} \exp\left(-\int_0^z \left(\sum_{j=1}^{n} k_j \omega_j(\sigma, t) \right) d\sigma - K_{bg} z \right) \tag{12}$$

where I_{in} is incident light intensity at the top of the water column, k_j is the specific light attenuation coefficient of the phytoplankton species j; K_{bg} is the background turbitiy (eg, not phytoplankton) and σ is an integration variable.

In code, we calculate that with the following function,

```
def self_shading(ListofAlgae, I_in, turbidity):
    ''' Use Lambert-Beer's Law to calculate light intensity at each depth '''
    corrected_z = (-1) * z # We want water depth [cm] to be zero
    # at the top, 20 at the bottom (positive numbers)
    kxC = np.zeros(N)
    for species in ListofAlgae:
        kxC += species.k * species.c

    I = np.zeros(N)
    vector = kxC - (turbidity*corrected_z)
    for i in (range(N)):
        I[i] = np.sum(vector[i:-1]) * z[i]

    I[I<0] = 1e-10
    I = I_in * np.exp(-I)
```

we can calculate the photic depth with,

```
# Get first element where light is less than 90% of I_in
photic_depth =  z[I<(0.9 * I_in)][-1]
```

Some examples of what algal growth looks like, using common parameters sourced from the Huisman paper, are shown below. In these examples there's no fluid motion at all (or particle sinking)– just looking purely at growth rate as a function of light. With constant light, we see that algal growth increases monotonically – within three days the population has increased over four-fold. Note that below the photic zone, however, algal are dying (albeit much more slowly). Thus, the net change in *total algal population* is determined by the *ratio of the photic zone* to the rest of the water column (assuming algae are dispersed uniformly). While growth may expload at the surface, if the rest of the algae are steadily dying, the net population could decline.

We can also impose a more-realistic light field which varies as a function of time (e.g., the diurnal light cycle.) Now, the algae are only exposed to light for 12 hours a day. They resulting figure shows a far more variable algal mass. If we sum the concentration over the water column, we see that mass is declining.

Combining growth with advection-diffusion

Finally, we can plug in our loss+growth equation with algae in the advection-diffusion model. Here are plots of velocity, κ_z, and algae concentration.

The TKE equation

We can derive an equation for the turbulent kinetic energy, q^2. This process is fairly intensive so I will put it in an appendix or something. The equation for the turbulent kinetic energy q^2 given by,

$$\underbrace{\frac{\partial q^2}{\partial t}}_{1} = \underbrace{\frac{\partial}{\partial z}(k_q \frac{\partial q^2}{\partial z})}_{2} + \underbrace{2P}_{2} + \underbrace{2B}_{3} - \underbrace{2\epsilon}_{4}$$

Moving from left to right,

$$[1] \quad \frac{\partial q^2}{\partial t}$$

is **unsteadiness**, or the change in turbulent kinetic energy over time;

$$[2] \quad \frac{\partial}{\partial z}(k_q \frac{\partial q^2}{\partial z})$$

is the **turbulent diffusion**, or a measure of how turbulent energy "diffuses" throughout the fluid flow. This neither creates nor destroys turbulence– just moves it. Next is

$$[3] \quad P = \overline{u_i u_k} \frac{\partial u_i}{\partial x_k},$$

production, also called **shear production**. We expand this for our simplified scenario (neglecting gradients of x, y, $w = 0$)

$$P = \overline{u'w'} \frac{\partial u}{\partial z} + \overline{v'w'} \frac{\partial v}{\partial z} + \overline{w'w'} \cancel{\frac{\partial w}{\partial z}}$$

$$P = \mu_t \left(\frac{\partial \overline{u}}{\partial z} + \cancel{\frac{\partial \overline{w}}{\partial z}} \right) \frac{\partial u}{\partial z} + \mu_t \left(\frac{\partial \overline{v}}{\partial z} + \cancel{\frac{\partial \overline{w}}{\partial y}} \right) \frac{\partial v}{\partial z}$$

$$P = \mu_t \left(\frac{\partial \overline{u}}{\partial z} \right)^2 + \mu_t \left(\frac{\partial \overline{v}}{\partial z} \right)^2$$

Dissipation is a disaster

$$[4] \quad \epsilon = -\nu \left(\overline{\frac{\partial u_i}{\partial x_k} \frac{\partial u_i}{\partial x_k}} \right)$$

$$\epsilon = -\nu \left(\overline{\left(\frac{\partial u}{\partial z} \right)^2} + \overline{\left(\frac{\partial v}{\partial z} \right)^2} + \overline{\left(\cancel{\frac{\partial w}{\partial z}} \right)^2} \right)$$

$$\frac{\partial q^2}{\partial t} = \frac{\partial}{\partial z}(k_q \frac{\partial q^2}{\partial z}) + \nu_t \frac{\partial u}{dz}^2 + 2 \frac{g}{\rho_0} \kappa_p \frac{\partial p}{\partial z} - 2 \frac{(q^2)^{3/2}}{B_1 l}$$

Similar to our simplications before, we can comfortably neglect advection of TKE as long as the time scale of dissipation is shorter than the advection time scale T_a over one grid cell in the horizontal.

$$\frac{\partial q^2}{\partial t} = \frac{\partial}{\partial z}(k_q \frac{\partial q^2}{\partial z}) - \overline{u_i u_k}\frac{\partial u_i}{\partial x_k} + 2B - \nu \overline{\frac{\partial u_i}{\partial x_k}\frac{\partial u_i}{\partial x_k}}$$

Boundary layer approximation for environmental flow

The equations used in our model assume that in our environment, the lateral length scales L_x and L_y are far greater than the vertical length scale (or depth) L_z. This means that gradients in the x and y directions can be seen as very small, or negligble. We maintain gradients in the pressure direction as these drive the flow and are assumed not to be functions of x.

$$\frac{\partial u}{\partial t} + u\cancel{\frac{\partial u}{\partial x}} + v\cancel{\frac{\partial u}{\partial y}} + w\frac{\partial z}{\partial x} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu_t \left(\cancel{\frac{\partial^2 u}{\partial x^2}} + \cancel{\frac{\partial^2 u}{\partial y^2}} + \frac{\partial^2 u}{\partial z^2} \right)$$

$$\frac{\partial u}{\partial t} + w\frac{\partial z}{\partial x} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu_t \left(\frac{\partial^2 u}{\partial z^2} \right)$$

We also can prove that $w = 0$, simplifying our equation to

$$\frac{\partial u}{\partial t} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu_t \left(\frac{\partial^2 u}{\partial z^2} \right) \tag{13}$$

Boussinesq gradient diffusion hypothesis

Boussinesq attempted to simplify the mess of the Reynold's stress tensor $\overline{u_i u_j}$ by reasoning that these correlation terms would act like a stress tensor in a flow.

In deriving the Navier-Stokes we define a stress tensor as

$$-(\tau_{ij} + P\delta_{ij}) = -2\mu S_{ij} = -\mu \left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i}) \right)$$

where τ_{ij} represents shear stresses and $P\delta_{ij}$, or pressure, acts as a body force along the diagonal of the tensor.

If the Reynold's stresses are like shear stresses imposed on flow, we can subsitute them in here

$$-(\overline{u_i u_j} + \frac{2}{3}k\delta_{ij}) = -\mu_t \left(\frac{\partial \overline{u_i}}{\partial x_j} + \frac{\partial \overline{u_j}}{\partial x_i} \right)$$

only this turbulent stress tensor would be approximated by some turbulent viscosity μ_t and the

$$\overline{u_i u_j} = \mu_t \left(\frac{\partial \overline{u_i}}{\partial x_j} + \frac{\partial \overline{u_j}}{\partial x_i} \right) - \frac{2}{3} k \delta_{ij}$$

Note that the trace of the RHS would be

$$\mu_t (2 \frac{\partial u_i}{\partial x_i} - \frac{2}{3} k) = \mu_T (-3 \frac{2}{3} k) = -2 \mu_T k = -2 \mu_T (\frac{1}{2} \overline{u_i' u_i'})$$

The main thing to note here is that using this approxmiation, we can now represent the Reynold's stress as being driven by gradients in the mean flow. This will help us from a closure perspective.

The two-equation model

Our equation for the turbulent kinetic energy q is now given by,

$$\frac{\partial q^2}{\partial t} = \frac{\partial}{\partial z} (k_q \frac{\partial q^2}{\partial z}) + 2P + 2B - 2\epsilon$$

$$\frac{\partial q^2}{\partial t} = \frac{\partial}{\partial z} (k_q \frac{\partial q^2}{\partial z}) - \overline{u_i u_k} \frac{\partial u_i}{\partial x_k} + 2B - \nu \frac{\partial u_i}{\partial x_k} \frac{\partial u_i}{\partial x_k}$$

$$\frac{\partial q^2}{\partial t} = \frac{\partial}{\partial z} (k_q \frac{\partial q^2}{\partial z}) + \nu_t \frac{\partial u}{dz}^2 + 2 \frac{g}{\rho_0} \kappa_p \frac{\partial p}{\partial z} - 2 \frac{(q^2)^{3/2}}{B_1 l}$$

We can neglect advection as long as the time scale of dissipation ir short than the advection time scale T_a over one grid cell in the horizontal.

The Mellor-Yamada 2.5 closure

The chosen closure scheme for our 1-D turbulent model is the Mellor-Yamada 2.5 scheme. This scheme approximates the eddy viscosity as

$$\nu_t = S_m q L \tag{14}$$

and the turbulent diffusivity as

$$\kappa_q = S_h Q L, \tag{15}$$

where S_m and S_h are empircally-derived stability functions, q is the turbulent kinetic energy, and l is our turbulent length scale.

$$\frac{\partial q^2}{\partial t} = \frac{\partial}{\partial z}\left(qlS_q\frac{\partial q^2}{\partial z}\right) + 2\left(P_s + P_b - \frac{q^3}{B_1 l}\right) \tag{16}$$

$$\frac{\partial q^2 l}{\partial t} = \frac{\partial}{\partial z}\left(qlS_q\frac{\partial q^2 l}{\partial z}\right) + lE_1(P_s + P_b) - \frac{q^3}{B_1 l}W(z) \tag{17}$$

where P_s denotes shear production, and is given by

$$P_s = \nu_T\left(\left(\frac{\partial U}{\partial z}\right)^2 + \left(\frac{\partial V}{\partial z}\right)^2\right),$$

the buoyancy production, P_b, is calculated as

$$P_b = -\kappa_t\frac{g}{\rho_0}\frac{\partial\overline{\rho}}{\partial z},$$

and finally the turbulent dissipation is given by

$$\epsilon = \frac{q^3}{B_1 l}.$$

The function $W(z)$ is a wall-proximity function that adjusts dissipation as z approaches a boundary. S_q and E_1 are both empirical constants.

Turbulent kinetic energy equation

$$\frac{\partial q^2}{\partial t} = \frac{\partial}{\partial z}(k_q\frac{\partial q^2}{\partial z}) + 2P + 2B - 2\epsilon$$

$$\frac{\partial q^2}{\partial t} = \frac{\partial}{\partial z}(k_q\frac{\partial q^2}{\partial z}) + \nu_t\frac{\partial u}{\partial z}^2 + 2\frac{g}{\rho_0}\kappa_p\frac{\partial p}{\partial z} - 2\frac{(q^2)^{3/2}}{B_1 l}$$

$$\frac{\partial q^2}{\partial t} = \frac{\partial}{\partial z}(k_q\frac{\partial q^2}{\partial z}) + \underbrace{\nu_t\left(\frac{\partial u}{\partial z}\right)^2}_{\text{shear production}} + \underbrace{2\frac{g}{\rho_0}\kappa_p\frac{\partial p}{\partial z}}_{\text{bouyancy production}} \underbrace{-2\frac{(q^2)^{3/2}}{B_1 l}}_{\text{dissipation}}$$

We will think of barotropic pressure is $f(t)$ and barolinic is $f(t,z)$. We will write this gradient $\frac{\partial p}{\partial x}$ as P_x, the pressure gradient in the x-direction. We can now think about discretizing this equation,

$$\frac{\partial q^2}{\partial t} = \underbrace{\frac{\partial}{\partial z}(k_q \frac{\partial q^2}{\partial z})}_{n+1} + \underbrace{\nu_t \left(\frac{\partial u}{\partial z}\right)^2}_{n} + \underbrace{2\frac{g}{\rho_0}\kappa_p \frac{\partial p}{\partial z}}_{n} - \underbrace{2\frac{(q^2)^{3/2}}{B_1 l}}_{\text{mixed}}$$

Start to discretize...($Q = q^2$)

$$\frac{Q_i^{n+1} - Q_i^n}{\Delta t} = \frac{1}{\Delta z}\left(k_q \frac{Q_i^{n+1} - Q_{i-1}^{n+1}}{\Delta z}\right) + (\nu_t)_i^n \left(\left(\frac{\partial u}{\partial z}\right)_i^n\right)^2 + 2\frac{g}{\rho_0}(\kappa_p)_i^n \frac{\partial p}{\partial z} - 2\frac{(q^2)^{3/2}}{B_1 l}$$

We write the dissipation as a mixed discretization,

$$-2\left(\frac{\sqrt{Q}}{B_1 l}\right)_i^n \cdot Q_i^{n+1}$$

where it's partially linearized and partially explicit.

We start with the continuity equation,

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_i}(\rho U_i) = 0 \tag{18}$$

where flow is considered to be divergent-free, or solenoidal.

Meanwhile, the Reynolds-Averaged Navier Stokes reads:

$$\frac{\partial \overline{U}}{\partial t} + \overline{U_j}\frac{\partial \overline{U_k}}{\partial x_j} + \rho\epsilon_{jkl} f_k U_l = \frac{\partial}{\partial x_k}\overline{u_k u_j} - \frac{\partial \overline{P}}{\partial x_j} - g\hat{e_3} \tag{19}$$

The Boussinesq approximation is invoked to represent that density fluctuaions affect bouyancy but no other transport/state terms. These fluctuations are written $\beta\theta$, where β is the coefficient of thermal expansion.

Mellor and Herring (1973) uses the following equation:

$$\langle\frac{p}{\rho}(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i})\rangle = -\frac{q}{3l_1}(\overline{u_i u_j} - \frac{\delta_{ij}}{3}q^2) + C_1 q^2(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i}) \tag{20}$$

We are establishing equations for the following: $\overline{u_k u_i u_j}$, $\overline{p u_j}$, $\overline{u_i u_j \theta}$, and $\overline{p\theta}$

Math Note The velocity gradient $\frac{\partial U_j}{\partial x_i}$ can be decomposed into symmetric and antisymmetric parths,

$$\frac{\partial U_j}{\partial x_i} = S_{ij} + \Omega_{ij}$$

where the symmetric rate-of-strain tensor is

$$S_{ij} = \frac{1}{2}\left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i}\right)$$

and the asymmetric part is,

$$\Omega_{ij} = \frac{1}{2}\left(\frac{\partial U_i}{\partial x_j} - \frac{\partial U_j}{\partial x_i}\right)$$

Parameterization

$$\langle u_k u_i u_j \rangle = \frac{3}{5}\ lqS_q \left(\frac{\partial \langle u_i u_j \rangle}{\partial x_k} + \frac{\partial \langle u_i u_k \rangle}{\partial x_j} + \frac{\partial \langle u_j u_k \rangle}{\partial x_i}\right) \tag{21}$$

In this equation S_q is a dimensionless number.

Sm :

$$\frac{A_1}{A_2}\frac{B_1(\Gamma - C_1) - (B_1(\Gamma_1 - C_1) + 6(A_1 + 3A_2))R_f}{B_1\Gamma_1 - (B_1(\Gamma_1 + \Gamma_2) - 3A_1)R_f} \tag{22}$$

where

$$\Gamma_1 = \frac{1}{3} - \frac{2A_1}{B_1}$$

and

$$\Gamma_2 = \frac{B_2}{B_1} - \frac{6A_1}{B_1}$$

Sh:

$$S_h = \frac{A_2(1 - \frac{6A_1}{B_1})}{1 - (3A_2B_2G_h + 18A_1A_2G_h)} = \frac{A_2(1 - \frac{6A_1}{B_1})}{(1 - 3A_2G_h(B_2 + 6A_1))}$$

$$S_m = \frac{A_1(1 - 3C_1 - \frac{6A_1}{B_1}) + S_HG_h(18A_1 + 9A_1A_2)}{(1 - 9A_1A_2G_h)}$$

Richardson's number:

$$G_H = -\frac{L^2}{Q^2}\frac{g}{\rho_0}\frac{\partial}{\partial z}$$

Chosen parameters

$$C_1 = \Gamma_1 - \frac{1}{3A_1} B_1^{1/3} \approx 0.08$$

$$B_2 = \frac{B_1^{1/3}}{P_{rt}} \frac{u_T^2 \langle \theta^2 \rangle}{H^2} \approx$$
$$B_1 = 16.6$$

writing equations based on code

$$bX = bX - \frac{aXcX}{bX}$$

$$dX = dX - \frac{aXdX}{bX}$$

$$u_* = U_{bed} \sqrt{C_d}$$

$$G_h = -\left(\frac{N_{BV} * L}{Q + Small} \right)^2$$

$$diss = 2 \frac{dt * \sqrt{Q^2}}{B_1 L_p}$$